

【8.23| Economist】 Well Informed 心里有数——健康专栏

难度评级: ★★★☆☆

【考研那些事儿之科技系列】

今日文章脉络:

肠道微生物的重要意义 (Para. 1)



微生物组多样性有助于保持肠道健康 (Para. 2)



饮食多样性益于肠道微生物多样性 (Para. 3)



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公司的微生物组检测并不准确 (Para. 7)



与其花钱检测，不如买罐泡菜拖延导致收入下降、医疗延误和身体疼痛 (Para. 8)

Can you **hack** your **gut microbiome**?

你能黑进自己的肠道微生物组吗？

【学习流程：第一遍请先看原文自己精读，第二遍对照着本精读文档进行理解。文章标注的单词为重点单词，只提供相关注释，想要详细释义麻烦同学查一下词典哦！另外文章底色为灰色的句子为今天的翻译作业，今日文档中的句子分析模块会提供答案，同学们请先思考再看句子分析~】

【Para.1】

The trillions of **bugs** in a human gut digest complex **carbohydrates** into fatty acids that provide about 10% of an individual's calories. They **synthesise** vitamins, notably K (which regulates blood **clotting**) and various members of the B group. They help **tune** the immune system. They **suppress** the growth of **pathogens**. And they regulate production of signalling molecules, such as **serotonin**, involved in sending messages from the gut to the brain.

- **hack v.** /hæk/ to find a way of looking at and/or changing information on sb else's computer system without permission 非法侵入（他人的计算机系统）；黑客攻击【考研大纲词汇】

The bank's security system was hacked by a group of cybercriminals. 该银行的安全系统遭到一群网络罪犯的非法侵入。

- **gut n.** /gʌt/ the tube in the body through which food passes when it leaves the stomach 消化道；肠道【考研大纲词汇】

The bacteria in your gut play a crucial role in digestion. 你肠道中的细菌在消化过程中起着至关重要的作用。

guts n. /gʌts/ (informal) the courage and determination that it takes to do sth difficult or unpleasant 勇气；胆量；决心；毅力

He doesn't have the guts to walk away from a well-paid job. 他没胆量辞去一份报酬优厚的工作。

- **microbiome n.** /ˌmaɪkrəʊˈbaɪəʊm/ the community of microorganisms (such as bacteria, fungi, and viruses) that live in a particular environment, especially in the human body 微生物组；微生物群落

The human microbiome is of particular interest, as these commensal bacteria are important for human health and immunity. 人类微生物组尤其令人关注，因为这些共生细菌对人类健康和免疫力至关重要。

- **bug** n. /bʌg/ (informal) an illness that is usually not serious and is caused by bacteria or a virus (引起小毛病的) 细菌；病毒【考研大纲词汇】

I had a tummy/stomach bug last week. 上周我的肠胃出了点小毛病。

- **carbohydrate** n. /ˌkɑːbəʊˈhaɪdreɪt/ a substance such as sugar or starch that consists of carbon, hydrogen and oxygen. Carbohydrates in food provide the body with energy and heat. 碳水化合物；糖类【考研大纲词汇】

Nuts are high in carbohydrates. 坚果富含碳水化合物。

- **synthesise** v. /ˈsɪnθəsaɪz/ (also **synthesize**) to produce a substance by a chemical reaction in plants or animals 合成【考研大纲词汇】

There are many vitamins that the body cannot synthesize itself. 有很多维生素是人体自身无法合成的。

ANT break down

- **clot** v. /klot/ when blood or cream clots or when sth clots it, it forms thick, almost solid masses (血液或奶油) 凝结成块；使凝结【考研大纲词汇】

He was rushed into hospital because his blood wasn't clotting properly. 他被匆忙送进医院，因为他的血液不能正常凝结。

n. an almost solid piece of something 凝块

He had a blood clot removed from his brain. 他颅内的一个血块被取了出来。

n. a stupid person 傻瓜，笨蛋

Look what you've done, you clot! 看看你干了什么，笨蛋！

- **tune v. /tju:n/** to prepare or change sth so that it is suitable for a particular situation 调整；使适合；使协调【考研大纲词汇】

His speech was tuned to what the audience wanted to hear. 他的演讲根据听众的期望进行了调整。

be in/out of tune with someone/something *idiom* if you are in tune with people or ideas, you understand or agree with them, and if you are out of tune with them, you do not. 与...协调 / 不协调，与...相 / 不相契合

Much of his success comes from being in tune with what his customers want. 他的成功很大程度上归功于他契合了顾客的需要。

Her theories were out of tune with the scientific thinking of the time. 她的理论与当时的科学思想不符。

- **suppress v.** /sə'pres/ to prevent something from being seen or expressed or from operating 抑制，忍住，压抑；查禁【考研大纲词汇】

The virus suppresses the body's immune system. 这种病毒会抑制人体的免疫系统。

后缀：press 按，压

compress v. /kəm'pres/ (被)压紧 (com- 共同，一起 + press 按，压)

depress v. /di'pres/ 使沮丧 (de- 离开，向下 + press 按，压)

oppress v. /ə'pres/ 压迫 (op- 逆，相反 + press 按，压)

- **pathogen n.** /'pæθədʒən/ any small organism, such as a virus or a bacterium that can cause disease 病原体【考研大纲词汇】

The spread of pathogens by insects. 病原体通过昆虫传播。

- **serotonin n.** /,serə'təʊnɪn/ a chemical in the brain that affects how messages are sent from the brain to the body, and also affects how a person feels 血清素【考研大纲词汇】

Serotonin—the happiness hormone. 血清素——快乐激素。

人体肠道中数以万亿计的微生物会将复杂碳水化合物消化分解成脂肪酸，为人体提供约 10%的热量。它们还能合成维生素，尤其是维生素 K（能够调节血液凝固）以及多种 B 族维生素；帮助调节免疫系统；抑制病原体的生长；并调控血清素等信号分子的生成，向大脑传递肠道信息也有这些信号分子的功劳。

【Para.2】

Keeping your gut microbiome in **tip-top** condition is thus wise. Maintaining a diverse mix of bacteria in it is widely thought a good way to do this. And, since different species have different **appetites**, diversity of diet makes sense.

- **tip-top adj.** /'tɪp.tɒp/ excellent; perfect 非常优秀的，完美的

Even though our house is in tip-top condition, we're having problems selling it. 虽然我们的房子状况很好，但现在把它卖出去还是有困难。

- **appetite n.** /'æpɪtaɪt/ the feeling that you want to eat food 食欲，胃口【考研大纲词汇】

Both viruses cause fever and loss of appetite. 两种病毒都会引起发烧以及食欲不振。

n. a strong desire for something 强烈欲望；渴望

Birds have different appetites for different types of insects.

不同鸟类对不同类型的昆虫有着不同的偏好。

因此，保持肠道微生物组处于最佳状态实属明智之举。人们普遍认为，维持肠道内菌群多样化有助于保证微生物群的良好状态。而且，由于不同菌种的口味各不相同，饮食种类丰富也十分重要。

【Para.3】

In 2018 an American citizen-science project showed that people eating 30 types of plants a week had significantly more diverse microbiomes than those eating fewer than ten. And work from Stanford University, published in 2021, confirmed that **fermented** foods such as kimchi and sauerkraut diversify microbiomes, too. Some people, though, feel such **off-the-peg** advice is **insufficient**. They would prefer something **bespoke**.

- **ferment v.** /fə'ment/ if food or drink ferments or if you ferment it, it goes through a chemical change because of the action of yeast or bacteria, which may cause it to produce bubbles or heat, or turn sugars in it into alcohol.

(使) 发酵【考研大纲词汇】

You make wine by leaving grape juice to ferment until all the sugar has turned to alcohol. 酿葡萄酒是让葡萄汁发酵，直到所有的糖分都转化为酒精。

- **off-the-peg adj.** /ˌɒf.ðəˈpeg/ (US off-the-rack); (Australian English off-the-hook) made and bought in standard sizes and not made especially to fit a particular person (衣服) 均码的，标准码的；非定制的

She bought an off-the-peg suit because she couldn't afford a tailored one. 她买了一套成品西装，因为她买不起定制的。

- **insufficient adj.** /ˌɪnsəˈfɪʃnt/ not enough 不够的，不足的；不充分的【考研大纲词汇】

The funds available were insufficient to cover the cost of the project. 可用的资金不足以支付该项目的费用。

SYN inadequate, deficient

ANT sufficient, adequate, ample

- **bespoke adj.** /bɪˈspəʊk/ specially made for a particular person 定做的；定制的

He wears bespoke suits made by a tailor in Savile Row. 他穿的是萨维尔街裁缝为他量身定制的西装。

2018 年，一项由美国公众参与的科学研究项目显示，每周食用 30 种植物食物的人，其肠道微生物组多样性明显高于每周食用不足 10 种的人。斯坦福大学 2021 年发表的一项研究也证实，泡菜、酸菜等发酵食品同样能够增加肠道微生物组的多样性。不过，有些人觉得这种一刀切式的建议还不够。他们更希望得到量身定制的方案。

【Para.4】

For a suitable **fistful** of dollars, they can have it. **So-called** direct-to-consumer (D2C) firms will analyse **stool** samples and **compile** lists of the bugs **therein**, based on the DNA and RNA they find. Such firms may also offer **tailored** dietary recommendations, doses of live bacteria and special bug-friendly **supplements**, all intended to fill any microbial blanks that their tests have **flagged up**.

- **fistful n.** /'fɪst.fʊl/ an amount of something that you can hold in your fist 一把（之量）

He held out a fistful of crumpled dollars. 他拿出一把皱巴巴的钞票。

- **so-called adj.** /,səʊ 'kɔ:ld/ used to show that you think a word that is used to describe sb/sth is not suitable or not correct 所谓的

Her so-called friends never helped her when she was in trouble. 她那些所谓的朋友在她遇到麻烦时从没帮过她。

adj. used to introduce a new word or phrase that is not yet known by many people 叫做...的, 名叫...的;所谓的

It isn't yet clear how dangerous these so-called "super-rats" are. 目前还不清楚这种名叫"超级老鼠"的东西有多危险。

- **stool n.** /stu:l/ a piece of solid waste from the body 粪便

He told the doctor he had been passing bloody stools. 他告诉医生他便血。

- **compile v.** /kəm'paɪl/ to collect information from different places and arrange it in a book, report, or list 汇编; 编纂; 编辑; 编制【考研大纲词汇】

We're compiling some facts and figures for a documentary on the subject. 我们正在为一部有关这个主题的纪录片汇编资料。

- **therein adv.** /,ðeər'ɪn/ in or into a particular place, thing, etc. 在那里;在其中;在那点上

It is a thrilling tale of a haunted house and the ghosts therein. 那是一个令人毛骨悚然的故事, 讲的是一座凶宅还有里面的幽灵。

- **tailored adj.** /'teɪləd/ (of clothes) made to fit a particular person by a tailor (衣服) 量身定制的, 合身的

He wore a beautifully tailored suit to the wedding. 他穿了一套剪裁精美的西装去参加婚礼。

SYN customized, personalized

- **supplement n.** /'sʌplɪmənt/ something that is added to something else in order to improve it or complete it; something extra 补充物, 增补物【考研大纲词汇】

Most doctors recommend taking vitamin supplements if you are pregnant. 大多数医生建议孕妇服用维生素补充剂。

v. /'sʌp.lɪ.ment/ to add something to something to make it larger or better 增加, 补充, 增补

Some vegetarians like to supplement their diets with iron. 有些素食主义者喜欢在饮食之外吃一些含铁的药片。

- **flag up phrasal verb** to draw attention to sth 引起对...的注意

The report flagged up the dangers of under-age drinking. 这篇报道引起大众对未成年人饮酒危害的关注。

只要花上一笔钱, 他们就能得到这样的服务。所谓直接面向消费者 (D2C) 的公司会分析粪便样本, 根据其中检测到的 DNA 和 RNA, 列出样本中所含微生物的清单。这类公司可能

还会提供个性化的饮食建议、活菌剂量以及专门有利于微生物生长的补充剂，旨在填补检测所发现的微生物缺失。

【Para.5】

The problem with this approach is that there is no "right" microbiome to act as a reference. In fact, microbiomes can differ quite a bit from each other while still being perfectly effective at delivering a desirable outcome to their host. The biggest red flag for poor gut health is a simple lack of biodiversity—which is what a varied and partly fermented diet is good at correcting. By contrast, the idea that tinkering with the presence or absence of this or that type of bug will perceptibly improve your health has scant published scientific basis, and many D2C firms are reluctant to share the details of proprietary data on which they base their claims.

- **tinker v.** /'tɪŋkə(r)/ to make small changes to something, especially in an attempt to repair or improve it 修补；改进；摆弄

He spent the afternoon tinkering with his bicycle chain. 他花了一下午摆弄他的自行车链条。

I wish the government would stop tinkering with the health service. 我希望政府不要再在公共医疗卫生服务上折腾了。

- **scant adj.** /skænt/ very little; not enough 很少的；不足的；欠缺的

The project received scant attention from the management. 该项目没有得到管理层足够的关注。

adj. almost 接近；差不多

Add a scant half litre of milk to the mixture and stir well. 把接近半升的牛奶加入到混合物中，并搅拌均匀。

- **proprietary adj.** /prə'praɪətri/ (especially of information, software, or a product) made and sold by a particular company and protected by a patent, trademark, or copyright from being copied or used by others (尤指信息、软件或产品) 专有的，专卖的，专利的

The company refuses to reveal the proprietary formula of its best-selling drink. 该公司拒绝透露其畅销饮料的专利配方。

这种方法的问题在于，可以作为参照标准的所谓“正确”微生物组并不存在。事实上，不同人体内的微生物组可能存在相当大的差异，却依然能够非常有效地为宿主带来理想的结果。肠道健康不佳最明显的警讯，是生物多样性不足，而多样化、且部分经过发酵的饮食恰恰有助于纠正这一问题。相比之下，认为通过调整某一种微生物的存在与否，就能让健康状况

得到明显改善，这一观点几乎没有什么已发表的科学依据；而且，许多 D2C 公司也不愿公开其相关主张所依据的专有数据的具体细节。

【Para.6】

There are exceptions. In Britain, Tim Spector has ridden the twin horses of academic science (a post at King's College, London) and **commerce** (he helped found a D2C firm called Zoe) without so far falling off. Results from Zoe's customers are fed, with their permission, into **respectable**, peer-reviewed research. So far, this suggests individually tailored diets do not have huge effects, but may help at the **margins**.

- **commerce n.** /'kɒmɜ:s/ the activity of buying and selling goods and services, especially in large amounts; trade 商业；贸易；（尤指大宗商品的）买卖【考研大纲词汇】

The new online platform has revolutionized cross-border commerce. 这个新的在线平台彻底改变了跨境贸易。

- **respectable adj.** /rɪ'spektəbl/ considered by society to be acceptable, good, or of sufficient quality; deserving respect and approval 体面的；值得尊敬的；相当好的；可接受的【考研大纲词汇】

The study was published in a respectable scientific journal.

该研究发表在一本相当权威的科学期刊上。

respectful adj. /rɪˈspektfəl/ showing admiration for someone or something 毕恭毕敬的；表示尊敬的；尊重的

"We're so pleased to meet you at last," he said in a respectful tone of voice. "我们终于和您见面了，真是万分高兴，"他毕恭毕敬地说道。

当然，也有例外。在英国，蒂姆·斯佩克特（Tim Spector）横跨学商两界，他在伦敦国王学院任职的同时，还参与创办了一家名为 Zoe 的 D2C 公司。到目前为止，他在两个身份之间来回切换，游刃有余。经客户许可，Zoe 公司收集到的用户数据会被纳入正规的同行评议研究。截至目前，这些研究表明，个性化饮食并不会产生特别显著的效果，但或许能在一些细微之处有所帮助。

【Para.7】

However, even if a bespoke approach does have value, it must surely depend on the **microbial censuses** carried out by D2C firms being **accurate**. To test that, researchers at America's National Institute of Standards and Technology prepared 21 sets of **homogenised** (and thus **identical**) faeces and sent three to each of seven D2C firms. When the results (which were published in February) came back there were notable

differences not only between different firms' analyses but also, in one case, between those of an individual company.

- **microbial adj.** /maɪ'krəʊbiəl/ relating to or caused by microbes (= very small living things, especially bacteria, that can cause disease) 微生物的；由微生物引起的

The research focuses on the microbial communities living in the human digestive system. 这项研究聚焦于生活在人类消化系统中的微生物群落。

- **census n.** /'sensəs/ a count for official purposes, especially one to count the number of people living in a country and to collect information about them （官方的）调查，统计；（尤指）人口普查【考研大纲词汇】

A national census is carried out every ten years in many countries. 许多国家每十年进行一次全国人口普查。

- **accurate adj.** /'ækjərət/ correct and true in every detail 准确的；精确的；正确无误的【考研大纲词汇】

The scientist needed accurate measurements to verify the experiment results. 这位科学家需要精确的测量数据来验证实验结果。

ANT inaccurate, incorrect, approximate, vague

- **homogenise v.** /hə'mɒdʒənaɪz/ to treat milk so that the cream is mixed into other parts of the liquid 使均质（尤指使牛奶中的油脂粒均匀分布）

The dairy industry homogenizes milk to prevent the cream from rising. 乳制品行业将牛奶均质化，以防止油脂漂浮上升。

v. to change something so that all its parts or features become the same or very similar （使）变均匀；（使）变纯一；（使）变类同

TV has homogenized the culture and language of large parts of the planet. 电视使地球上大部分地区的文化和语言同质化。

- **identical adj.** /aɪ'dentɪkl/ exactly the same; not different in any way 完全相同的；一模一样的【考研大纲词汇】

The two samples produced almost identical results in the laboratory test. 这两个样本在实验室测试中得出了几乎完全相同的结果。

不过，即便量身定制的方法确实有价值，它显然也必须建立在 D2C 公司所进行的微生物普查足够准确的基础上。为了检验这一点，美国国家标准与技术研究院的研究人员制备了 21 份经过均质处理、因而完全相同的粪便样本，并给向 7 家 D2C 公司每家寄出了 3 份。检测结果（已于 2 月发表）寄回后，人们

发现，不仅不同公司的分析结果之间存在明显差异，甚至有一家公司的检测结果内部也有不同。

【Para.8】

As Jonathan Eisen, an evolutionary microbiologist at the University of California, Davis, observes, just because microbiomes differ does not mean an era of personalised microbiome medicine is coming any time soon. In fact, he **puts** it more strongly than this, saying, "Most of the claims [in this area] are **bogus**." Perhaps better to save your money and invest in a jar of kimchi.

- **put v.** /pʊt/ to express or state something in a particular way 说；表达；表述

To put it simply, the experiment failed because of human error. 简单地说，实验失败是由于人为失误。

- **bogus adj.** /'bəʊgəs/ (especially of a claim or a document) not genuine; pretending to be real or true （尤指说法或文件）假的；伪造的；冒充的【考研大纲词汇】

The insurance company suspected the claim was bogus and launched an investigation. 该保险公司怀疑这项索赔是伪造的，并展开了调查。

加州大学戴维斯分校的进化微生物学家乔纳森·艾森 (Jonathan Eisen) 指出，微生物组存在差异，并不意味着个性化微生物组医学时代就会很快到来。其实，他本人的话更为直接：“（这个领域的）大多数说法都是骗人的。”所以，你也许还是省下这笔钱，买罐泡菜更好。

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今日翻译作业

(Para. 5) By contrast, the idea that tinkering with the presence or absence of this or that type of bug will perceptibly improve your health has scant published scientific basis, and many D2C firms are reluctant to share the details of proprietary data on which they base their claims. (下一页提供句子分析)

今日句子分析

Para. 5 By contrast, the idea that tinkering with the presence or absence of this or that type of bug will perceptibly improve your health has scant published scientific basis, and many D2C firms are reluctant to share the details of proprietary data on which they base their claims. 相比之下，认为通过调整某一种微生物的存在与否，就能让健康状况得到明显改善，这一观点几乎没有什么已发表的科学依据；而且，许多 D2C 公司也不愿公开其相关主张所依据的专有数据的具体细节。

主句 1

By contrast, the idea

伴随状语

主语

同位语从句

that tinkering with the presence or absence

主语

of this or that type of bug

后置定语

will perceptibly improve your health

程度状语

谓语

宾语

has scant published scientific basis,

谓语

宾语

主句 2

and many D2C firms are reluctant

主语

系语

表语

to share the details of proprietary data

表语补足语

定语从句（修饰 data）

on which they base their claims.

地点状语

主语

谓语

宾语

主句 1:

伴随状语：By contrast; (by contrast 对比前文某种比较有科学依据、看起来具有健康改善作用的观点/证据。而本句开始转向：作者认为，通过调整微生物来改善健康缺乏科学依据。注意，however 也可以用于对比，区别在于 however 表示转折，而 by contrast 表示“相比之下”，更强调比较两个对象之间的差异。)

主语：the idea;

同位语从句：（解释 idea 的内容。从句是主谓宾结构，已经是一个完整的句子，that 没有在里面充当任何成分，所以它不是定语从句。）

引导词：that; 主语：tinkering with the presence or absence;

(tinker with 不是简单的 change，而是对某物进行小幅调整、改动，带有“试着改改看”“随意调整”的意味。)

后置定语：of this or that type of bug; (this or that 在这里指的
的不是前文已有内容，不表示“这个或那个”，而是表示“某一种
或另一种”，简言之就是“某种”。如 You can choose this or
that option. 你可以选择某个选项。

the presence or absence (存在或不存在) of this or that type
of bug (某一种微生物的)，字面意思是“某一种微生物的存在或
不存在”，太啰嗦。译文简化为“某一种微生物的存在与否”。)

程度状语：perceptibly；谓语：will improve；宾语：your
health；谓语：has；宾语：scant published scientific basis；

主句 2：

连词：and；主语：many D2C firms；系语：are；表语：
reluctant；(and 并列两个分句，分句 1：the idea ... has scant
published scientific basis，分句 2：many D2C firms are
reluctant to share...。这句话的结构就是[A has B], and [C are
reluctant to D].)

表语补足语：to share the details of proprietary data；

定语从句（修饰 data）：

地点状语：on which；主语：they；谓语：base；宾语：their
claims；(their 指 many D2C firms)

写作积累

1. (Para. 6) So far, this suggests individually tailored diets do not have huge effects, but may help **at the margins**.

at the margins 表示 “being at the outer edge, border, or boundary of a group, place, or system, often used to describes people, ideas, or activities that have little power, influence, or involvement in the main part of society” , 即 “只在有限程度上、在细微之处, 仅带来边际性的” , 例如, Individually tailored diets may make a difference at the margin, even if their overall impact is limited. 即使个性化饮食的总体影响有限, 它在某些细微方面仍可能产生一定作用。此外, 还有以下表达可以表示类似的含义:

- ♦ **yield modest gains** *idiom* to produce small, limited, or average improvements, profits, or increases rather than large or dramatic breakthroughs 获得轻微的收益, 收效不大
The stock market recovery will yield modest gains for investors this year. 今年的股市复苏只会为投资者带来适度收益。
- ♦ **move the needle** *idiom* to make a noticeable, measurable difference or progress toward a goal 带来可感知的变化 (与 at the margins 并不是完全同义, 而是反向强调) , 产生足以被注意到的变化

The treatment barely moves the needle. 这种治疗几乎没有产生什么实质性改变。

2. (Para. 7) However, even if a bespoke approach does have value, it must surely depend on the microbial censuses **carried out** by D2C firms being accurate.

Carry out 表示 “to perform, complete, or execute a task, plan, order, or promise” , 即 “实施、开展、进行 (某项工作、研究、调查、检测等) ” , 例如, It is hard to carry out the new plan in a short time. 在短时间内落实这个新计划很难。此外, 还有以下表达可以表示类似的含义:

- ♦ **put something in place** *idiom* to set up, establish, or implement a system, rule, plan, or physical object so it is ready to use 实施, 使就位

The government put a new tax policy in place this year. 政府今年出台了一项新的税收政策。

- ♦ **go through with something** *idiom* to do a planned or promised action, especially when it is hard, scary, or you feel unsure about it 把...真正做下去, 执行到底

We have decided to go through with the original plan despite the high cost. 尽管代价高昂, 我们还是决定按原计划把事情进行到底。